

The stated ceiling rank bound is not sharp at two constraints

ziangni-sys

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We disprove the pointwise sharpness of the ceiling formula in conjecture 00000008822. The displayed rank inequality and the asserted attainment of the ceiling bound are incompatible at $m = 2$. We do not use the separate assertions about minimal-rank solutions or Frank–Wolfe.

For real symmetric semidefinite programs, write the actual feasible set as

$$F(A, b) = \{X \succeq 0 : \text{tr}(A_i X) = b_i, \ i = 1, \dots, m\}.$$

The conjecture asserts that the rank r of any extreme point satisfies

$$\frac{r(r+1)}{2} \leq m,$$

and claims that the bound $c(m) = \lceil (\sqrt{8m+1} - 1)/2 \rceil$ is tight: an extreme point can have rank exactly $c(m)$.

At $m = 2$, we have $3 < \sqrt{17} < 5$, hence

$$1 < \frac{\sqrt{17} - 1}{2} < 2, \quad c(2) = 2.$$

But an extreme point of that rank would, by the simultaneously asserted rank inequality, satisfy

$$3 = \frac{2(2+1)}{2} \leq 2,$$

a contradiction. More generally, for integer $r \geq 0$ the displayed inequality at $m = 2$ forces $r \leq 1$. This argument covers every dimension, all choices of the two trace constraints and every actual extreme point. Thus no such instance can attain the stated ceiling value under the conjecture's own rank law.

The correct integer upper bound obtained from the displayed inequality uses the *floor*, not the ceiling:

$$r \leq \left\lfloor \frac{\sqrt{8m+1} - 1}{2} \right\rfloor.$$

The ceiling is a weaker bound; calling it pointwise sharp is the error. For triangular values $m = r(r+1)/2$, floor and ceiling coincide. We do not deny that examples can attain the formula at those special values; the counterexample concerns its asserted sharpness as a bound in m .

A concrete feasible example at $m = 2$ is the scalar SDP with $A_1 = A_2 = [1]$ and $b_1 = b_2 = 1$. Its feasible set is the singleton $\{[1]\}$, whose sole point is genuinely extreme and has rank 1. This illustrates the attainable rank without replacing the universal contradiction by an example-specific assumption.

Formal correspondence. Lean defines $F(A, b)$ using the actual `Matrix.PosSemidef`, matrix multiplication and trace. It uses the actual `Set.extremePoints` and `Matrix.rank`, quantifies over all finite matrix dimensions and all two-constraint instances, computes $c(2) = 2$, and disproves the conjunction of the displayed universal rank law with ceiling attainment. It also checks the explicit singleton SDP and its rank-one extreme point. The inequality itself is a hypothesis of the source conjunction being refuted; the Lean proof does not pretend to prove Pataki's theorem. Lean 4.19.0 and Mathlib commit c44e0c8ee63ca166450922a373c7409c5d26b00b are pinned.